AWS 50 Services

Real-Time Hands-on Training Curriculum

Practical tasks covering configuration, deployment, integration, monitoring, security, and troubleshooting across 50 AWS services.

Training material

AWS Services

1. IAM
2. EC2
3. ELB
4. ASG
5. EBS
6. EFS
7. CloudFront
8. S3
9. Amazon Route53
10. ACM
11. Lambda
12. API Gateway
13. ECS
14. ECR
15. EKS
16. CloudWatch
17. SNS
18. SQS
19. SES
20. SSM
21. RDS
22. CloudTrail
23. KMS
24. AWS Secrets Manager
25. WAF
26. Amazon GuardDuty
27. AWS Security Hub
28. Organizations
29. Control Tower
30. Config
31. Amazon EventBridge
32. Athena
33. Trusted Advisor
34. Application Migration Service
35. Database Migration Service
36. Application Discovery Service
37. DataSync
38. Inspector
39. CodeBuild

40. CodeDeploy
41. CodePipeline
42. Health Dashboard
43. Cost Explorer
44. AWS Budgets
45. Amplify
46. Macie
47. DynamoDB
48. Aurora
49. VPC
50. AWS Transit Gateway

AWS 50 Services – Real-Time Hands-on Training Curriculum

1. AWS IAM (Identity and Access Management)

1. Create IAM users, groups, roles, and policies.
2. Create custom IAM policies using the JSON policy editor.
3. Attach AWS-managed and customer-managed policies to IAM users and groups.
4. Configure MFA for IAM users.
5. Reset IAM user console passwords and rotate access keys.
6. Configure IAM users to access AWS through Console, CLI, and SDK.
7. Configure IAM roles and assume roles using AWS STS.
8. Configure cross-account IAM role access between two AWS accounts.
9. Implement least-privilege access using IAM policies.
10. Configure IAM policy conditions based on IP address, region, and MFA.
11. Configure EC2 instance profiles and attach IAM roles to EC2.
12. Troubleshoot IAM AccessDenied errors using policy simulation and CloudTrail.

2. Amazon EC2 (Elastic Compute Cloud)

1. Launch Linux and Windows EC2 instances.
2. Configure key pairs and connect to instances using SSH and RDP.
3. Configure security groups and network access.
4. Change EC2 instance types (vertical scaling).
5. Stop, start, reboot, and terminate EC2 instances.
6. Create and manage AMIs from existing EC2 instances.
7. Launch new EC2 instances using custom AMIs.
8. Configure EC2 user data to install Apache, Nginx, and other packages automatically.
9. Configure EC2 instance metadata and retrieve instance information using IMDSv2.
10. Attach and detach EBS volumes.
11. Configure EC2 monitoring using CloudWatch.
12. Configure EC2 instance roles using IAM.
13. Configure EC2 Auto Recovery.

14. Troubleshoot EC2 status check failures and SSH connectivity issues.

15. Configure EC2 instances in public and private subnets.

3. Elastic Load Balancing (ELB)

1. Create an Application Load Balancer (ALB).

2. Create a Network Load Balancer (NLB).

3. Create target groups and register EC2 instances.

4. Configure health checks for target groups.

5. Configure HTTP and HTTPS listeners.

6. Configure host-based routing using multiple domains.

7. Configure path-based routing for different applications.

8. Configure SSL/TLS certificates using ACM.

9. Configure load balancing across multiple Availability Zones.

10. Configure ALB with Auto Scaling Groups.

11. Configure sticky sessions.

12. Configure ALB access logs and monitor requests.

13. Troubleshoot unhealthy targets, 502 errors, and 504 errors.

4. Amazon EC2 Auto Scaling (ASG)

1. Create an Auto Scaling Group.

2. Create and configure launch templates.

3. Configure minimum, maximum, and desired capacity.

4. Configure scaling policies based on CPU utilization.

5. Configure target tracking scaling policies.

6. Configure scheduled scaling policies.

7. Configure step scaling policies.

8. Integrate Auto Scaling Groups with Application Load Balancers.

9. Configure health checks and automatic instance replacement.

10. Configure scaling across multiple Availability Zones.

11. Test automatic scaling by generating CPU load.

12. Configure instance termination policies and lifecycle hooks.

5. Amazon EBS (Elastic Block Store)

1. Create and attach EBS volumes to EC2 instances.

2. Identify EBS devices using Linux commands.

3. Create partitions and format volumes using ext4 and XFS.

4. Mount EBS volumes and configure persistent mounts using `/etc/fstab`.

5. Extend existing EBS volumes.

6. Expand Linux partitions and file systems after increasing volume size.

7. Create EBS snapshots.

8. Restore EBS volumes from snapshots.

9. Copy EBS snapshots across AWS regions.

10. Configure EBS encryption using AWS KMS.

11. Configure EBS volume types: gp3, io2, st1, and sc1.

12. Troubleshoot EBS mounting and permission issues.

6. Amazon EFS (Elastic File System)

1. Create an EFS file system.

2. Configure EFS mount targets across multiple Availability Zones.

3. Configure security groups for EFS access.

4. Install and configure EFS utilities on EC2.

5. Mount EFS on Linux EC2 instances.

6. Configure persistent EFS mounting using `/etc/fstab`.

7. Mount the same EFS file system on multiple EC2 instances.

8. Configure EFS access points.

9. Configure IAM authorization for EFS.

10. Configure EFS encryption at rest and in transit.

11. Configure EFS lifecycle management and storage classes.

12. Troubleshoot EFS mount failures and access-denied errors.

7. Amazon CloudFront

1. Create a CloudFront distribution.

2. Configure an S3 bucket as an origin.

3. Configure an EC2 instance or ALB as an origin.

4. Configure Origin Access Control (OAC) for private S3 content.

5. Configure custom domain names using Route 53.

6. Integrate ACM certificates with CloudFront.

7. Configure HTTP-to-HTTPS redirection.

8. Configure cache behaviors and cache policies.

9. Configure path-based origin routing.

10. Create CloudFront cache invalidations.

11. Configure signed URLs and signed cookies.

12. Configure CloudFront geographic restrictions.

13. Enable CloudFront access logging and monitoring.

14. Troubleshoot 403, 404, and 502 errors.

8. Amazon S3 (Simple Storage Service)

1. Create and configure S3 buckets.

2. Upload, download, copy, and delete objects.

3. Configure bucket policies and IAM-based access.

4. Configure bucket versioning.

5. Enable static website hosting.

6. Configure lifecycle rules to transition objects between storage classes.

7. Configure S3 Intelligent-Tiering.

8. Configure server-side encryption using SSE-S3 and SSE-KMS.
9. Configure default bucket encryption.
10. Configure cross-region replication (CRR).
11. Configure same-region replication (SRR).
12. Configure S3 event notifications with Lambda and SNS.
13. Configure pre-signed URLs for secure file sharing.
14. Configure S3 Object Lock and retention policies.
15. Configure access logging and CloudTrail data events.
16. Host a static website using S3 and CloudFront.
17. Configure CORS for S3 buckets.
18. Troubleshoot AccessDenied and bucket policy issues.

9. Amazon Route 53

1. Create public hosted zones.
2. Create private hosted zones.
3. Configure A, AAAA, CNAME, MX, TXT, and NS records.
4. Configure domain-to-EC2 IP mapping.
5. Configure alias records pointing to ALB and CloudFront.
6. Configure domain names for applications using Route 53.
7. Configure health checks.
8. Configure failover routing policies.
9. Configure weighted routing policies.
10. Configure latency-based routing.
11. Configure geolocation and geoproximity routing.
12. Configure DNS records for primary and secondary endpoints.
13. Troubleshoot DNS resolution using `nslookup` and `dig`.

10. AWS Certificate Manager (ACM)

1. Request a public SSL/TLS certificate.
2. Request a private certificate using ACM Private CA.
3. Validate domain ownership using DNS validation.
4. Validate domain ownership using email validation.
5. Configure wildcard certificates.
6. Attach ACM certificates to ALB.
7. Attach ACM certificates to CloudFront.
8. Configure HTTPS listeners using ACM certificates.
9. Configure automatic certificate renewal.
10. Monitor certificate expiration and validation status.
11. Troubleshoot certificate validation and domain mismatch issues.

11. AWS Lambda

1. Create Lambda functions using Python.

2. Create Lambda functions using Node.js.
3. Configure Lambda execution roles using IAM.
4. Configure environment variables.
5. Configure Lambda layers.
6. Upload deployment packages and dependencies.
7. Configure Lambda triggers using S3 events.
8. Integrate Lambda with API Gateway.
9. Integrate Lambda with SNS and SQS.
10. Configure Lambda with EventBridge schedules.
11. Configure Lambda VPC access to private resources.
12. Configure Lambda versions and aliases.
13. Configure reserved and provisioned concurrency.
14. Monitor Lambda execution using CloudWatch Logs and Metrics.
15. Configure Lambda Dead Letter Queues and destinations.
16. Troubleshoot timeout, memory, permission, and execution errors.
17. Build an automated EC2 start/stop solution using Lambda.

12. Amazon API Gateway

1. Create REST APIs.
2. Create HTTP APIs.
3. Create API resources and methods.
4. Integrate API Gateway with Lambda.
5. Integrate API Gateway with HTTP backends.
6. Configure GET, POST, PUT, and DELETE methods.
7. Configure request and response mappings.
8. Configure API keys and usage plans.
9. Configure throttling and rate limiting.
10. Configure CORS.
11. Configure authentication using IAM authorization.
12. Configure Amazon Cognito authorizers.
13. Configure custom domain names and ACM certificates.
14. Deploy APIs using different stages.
15. Configure stage variables.
16. Enable API Gateway access and execution logging.
17. Test APIs using Postman and curl.
18. Troubleshoot 403, 404, 429, and 500 errors.

13. Amazon ECS (Elastic Container Service)

1. Create an ECS cluster.
2. Deploy ECS applications using the EC2 launch type.
3. Deploy ECS applications using the Fargate launch type.

4. Create task definitions.
5. Configure CPU and memory limits.
6. Create and run ECS tasks.
7. Create ECS services.
8. Deploy Docker containers from ECR.
9. Configure ECS networking using public and private subnets.
10. Configure ECS task execution roles and task roles.
11. Configure Application Load Balancer integration.
12. Configure ECS service auto scaling.
13. Configure environment variables and secrets.
14. Mount EFS volumes into ECS containers.
15. Configure CloudWatch logging for ECS tasks.
16. Configure ECS rolling deployments.
17. Configure ECS blue/green deployments.
18. Troubleshoot ECS task startup failures, image pull failures, and container health checks.

14. Amazon ECR (Elastic Container Registry)

1. Create private ECR repositories.
2. Authenticate Docker with ECR.
3. Build Docker images locally.
4. Tag Docker images with ECR repository URLs.
5. Push Docker images to ECR.
6. Pull images from ECR onto EC2.
7. Configure ECR repository policies.
8. Configure image scanning.
9. Configure image tag immutability.
10. Configure lifecycle policies to delete older images.
11. Configure cross-account ECR access.
12. Configure cross-region image replication.
13. Integrate ECR with ECS and EKS.
14. Troubleshoot Docker authentication and image push/pull errors.

15. Amazon EKS (Elastic Kubernetes Service)

1. Create an EKS cluster.
2. Configure the AWS CLI, kubectl, and eksctl.
3. Create and configure managed node groups.
4. Configure Fargate profiles.
5. Deploy Kubernetes pods.
6. Create Kubernetes deployments and services.
7. Configure ClusterIP, NodePort, and LoadBalancer services.
8. Deploy applications using YAML manifests.

9. Deploy applications using Helm charts.
10. Integrate EKS with ECR.
11. Configure AWS Load Balancer Controller.
12. Configure IAM Roles for Service Accounts (IRSA).
13. Configure EKS Pod Identity.
14. Configure persistent storage using EBS CSI and EFS CSI drivers.
15. Configure Horizontal Pod Autoscaler.
16. Configure Cluster Autoscaler or Karpenter.
17. Configure EKS logging and monitoring.
18. Upgrade EKS cluster versions and node groups.
19. Troubleshoot CrashLoopBackOff, ImagePullBackOff, and networking issues.

16. Amazon CloudWatch

1. Create CloudWatch dashboards.
2. Create metric alarms for EC2 CPU utilization.
3. Monitor memory and disk utilization using CloudWatch Agent.
4. Configure CloudWatch Logs.
5. Create log groups and log streams.
6. Configure EC2 and application log collection.
7. Create metric filters from application logs.
8. Configure SNS notifications for CloudWatch alarms.
9. Create composite alarms.
10. Configure CloudWatch Logs Insights queries.
11. Configure custom metrics using AWS CLI and SDK.
12. Configure log retention policies.
13. Monitor Lambda, ECS, ALB, and RDS metrics.
14. Create CloudWatch Synthetics canaries.
15. Troubleshoot application performance using logs and metrics.

17. Amazon SNS (Simple Notification Service)

1. Create standard SNS topics.
2. Create FIFO SNS topics.
3. Subscribe email endpoints to SNS topics.
4. Subscribe HTTP/HTTPS endpoints.
5. Configure SMS notifications.
6. Configure SQS subscriptions to SNS.
7. Configure Lambda subscriptions to SNS.
8. Publish messages using AWS Console and CLI.
9. Configure SNS access policies.
10. Configure message filtering policies.
11. Configure delivery retries and dead-letter queues.

12. Build CloudWatch alarm notifications using SNS.

18. Amazon SQS (Simple Queue Service)

1. Create standard SQS queues.
2. Create FIFO SQS queues.
3. Configure message retention periods.
4. Configure visibility timeout.
5. Send, receive, and delete messages.
6. Configure long polling.
7. Configure message delay.
8. Configure dead-letter queues.
9. Configure redrive policies.
10. Integrate SQS with Lambda.
11. Integrate SQS with SNS using fan-out architecture.
12. Configure message encryption using KMS.
13. Configure queue access policies.
14. Implement message processing using Python and boto3.
15. Troubleshoot duplicate message processing and visibility timeout issues.

19. Amazon SES (Simple Email Service)

1. Create and verify an SES identity.
2. Verify an email address and domain.
3. Configure domain authentication using SPF, DKIM, and DMARC.
4. Move SES from sandbox to production.
5. Send emails using the AWS Console.
6. Send emails using AWS CLI.
7. Send emails using Python and boto3.
8. Configure HTML email templates.
9. Configure configuration sets.
10. Configure bounce and complaint notifications using SNS.
11. Configure email event publishing.
12. Troubleshoot email delivery and reputation issues.

20. AWS Systems Manager (SSM)

1. Configure SSM Agent on EC2 instances.
2. Configure IAM instance profiles for Systems Manager.
3. Connect to EC2 using Session Manager without SSH.
4. Execute shell commands using Run Command.
5. Execute commands on multiple EC2 instances.
6. Create and execute custom SSM documents.
7. Configure Patch Manager for Linux and Windows.
8. Configure Maintenance Windows.

9. Store configuration values using Parameter Store.
10. Store SecureString parameters using KMS.
11. Configure State Manager associations.
12. Configure Automation runbooks.
13. Use Fleet Manager to manage EC2 instances.
14. Configure Inventory collection.
15. Troubleshoot instances that are not appearing as managed nodes.

21. Amazon RDS (Relational Database Service)

1. Create MySQL and PostgreSQL RDS instances.
2. Configure database credentials and master passwords.
3. Configure DB subnet groups.
4. Configure security groups for database connectivity.
5. Connect to RDS from EC2 and local systems.
6. Create databases, tables, and database users.
7. Configure automated backups.
8. Create manual DB snapshots.
9. Restore databases from snapshots.
10. Modify DB instance types and storage.
11. Enable storage autoscaling.
12. Configure Multi-AZ deployments.
13. Create read replicas.
14. Configure encryption at rest using KMS.
15. Enable Enhanced Monitoring and Performance Insights.
16. Configure RDS logs and CloudWatch alarms.
17. Configure maintenance windows and backup retention.
18. Troubleshoot connection timeouts, authentication, and database performance issues.

22. AWS CloudTrail

1. Create and configure a CloudTrail trail.
2. Enable management event logging.
3. Configure read and write event selection.
4. Configure S3 as a CloudTrail log destination.
5. Enable CloudTrail log file validation.
6. Configure multi-region trails.
7. Enable CloudTrail data events for S3 and Lambda.
8. Query CloudTrail events using Event History.
9. Identify who created, modified, or deleted AWS resources.
10. Integrate CloudTrail with CloudWatch Logs.
11. Configure EventBridge rules for specific API activities.
12. Investigate suspicious API activity using CloudTrail.

23. AWS KMS (Key Management Service)

1. Create customer-managed KMS keys.
2. Create symmetric and asymmetric keys.
3. Configure key administrators and key users.
4. Configure key policies.
5. Configure key grants.
6. Encrypt and decrypt data using KMS.
7. Enable automatic key rotation.
8. Configure manual key rotation.
9. Encrypt S3 objects using SSE-KMS.
10. Encrypt EBS volumes using KMS.
11. Encrypt RDS databases using KMS.
12. Configure cross-account KMS access.
13. Schedule key deletion and cancel deletion.
14. Monitor KMS API activity using CloudTrail.

24. AWS Secrets Manager

1. Create secrets for database credentials.
2. Store application API keys and passwords.
3. Retrieve secrets using AWS CLI.
4. Retrieve secrets using Python and boto3.
5. Configure automatic secret rotation.
6. Configure secret rotation using Lambda.
7. Integrate Secrets Manager with RDS.
8. Integrate Secrets Manager with ECS task definitions.
9. Configure resource-based policies.
10. Configure cross-account secret access.
11. Configure secret replication across regions.
12. Monitor secret access using CloudTrail.

25. AWS WAF (Web Application Firewall)

1. Create a Web ACL.
2. Associate WAF with an Application Load Balancer.
3. Associate WAF with CloudFront.
4. Configure AWS Managed Rule Groups.
5. Configure IP-based blocking rules.
6. Create custom rules based on URI paths.
7. Configure rate-based rules to prevent excessive requests.
8. Configure geographic match rules.
9. Configure SQL injection protection.
10. Configure Cross-Site Scripting (XSS) protection.

11. Configure CAPTCHA and challenge actions.
12. Enable WAF logging.
13. Analyze blocked and allowed requests.
14. Troubleshoot false positives and rule priority issues.

26. Amazon GuardDuty

1. Enable GuardDuty in an AWS account.
2. Enable GuardDuty across multiple regions.
3. Configure GuardDuty delegated administrator using Organizations.
4. Review security findings.
5. Investigate suspicious EC2 network activity.
6. Investigate compromised IAM credentials.
7. Configure GuardDuty S3 Protection.
8. Configure EKS Protection.
9. Configure Malware Protection for EC2.
10. Integrate GuardDuty findings with EventBridge.
11. Configure SNS notifications for critical findings.
12. Generate sample findings and test the response workflow.
13. Archive and suppress findings using filters.

27. AWS Security Hub

1. Enable AWS Security Hub.
2. Enable security standards such as AWS Foundational Security Best Practices.
3. Enable CIS AWS Foundations Benchmark where supported.
4. Review security findings and compliance status.
5. Configure security controls.
6. Configure custom insights.
7. Integrate Security Hub with GuardDuty.
8. Integrate Security Hub with Inspector and Macie.
9. Configure cross-region aggregation.
10. Configure delegated administration using Organizations.
11. Automate finding responses using EventBridge and Lambda.
12. Generate sample findings and validate security workflows.

28. AWS Organizations

1. Create an AWS Organization.
2. Invite existing AWS accounts.
3. Create member accounts.
4. Organize accounts using Organizational Units (OUs).
5. Create Service Control Policies (SCPs).
6. Restrict access to selected AWS regions.
7. Deny access to specific AWS services using SCPs.

8. Configure centralized billing.
9. Configure consolidated billing reports.
10. Enable trusted access for supported AWS services.
11. Configure delegated administrator accounts.
12. Test SCP restrictions using member accounts.
13. Understand the difference between IAM policies and SCPs.

29. AWS Control Tower

1. Understand Control Tower architecture and landing zones.
2. Set up an AWS Control Tower landing zone.
3. Configure the home region and additional governed regions.
4. Create accounts using Account Factory.
5. Configure Organizational Units.
6. Enable preventive guardrails.
7. Enable detective guardrails.
8. Enable proactive controls where supported.
9. Monitor account compliance.
10. Configure account-level governance.
11. Integrate Control Tower with AWS Organizations.
12. Troubleshoot landing zone and account enrollment failures.

30. AWS Config

1. Enable AWS Config.
2. Configure configuration recorders.
3. Configure delivery channels using S3 and SNS.
4. View resource configuration history.
5. View resource relationships.
6. Enable AWS-managed Config rules.
7. Create custom Config rules using Lambda.
8. Detect publicly accessible S3 buckets.
9. Detect EC2 instances without required tags.
10. Detect unencrypted EBS volumes.
11. Configure automatic remediation using SSM Automation.
12. Create conformance packs.
13. Generate compliance reports.
14. Troubleshoot Config recording and rule evaluation issues.

31. Amazon EventBridge

1. Create EventBridge event buses.
2. Create custom event buses.
3. Create event rules for EC2 state changes.
4. Create scheduled rules for recurring tasks.

5. Configure event patterns using JSON.
6. Integrate EventBridge with Lambda.
7. Integrate EventBridge with SNS and SQS.
8. Configure cross-account event delivery.
9. Configure event archiving and replay.
10. Configure dead-letter queues.
11. Configure API destinations.
12. Build automated EC2 start/stop workflows.
13. Troubleshoot event pattern matching and failed target delivery.

32. Amazon Athena

1. Create Athena workgroups.
2. Configure S3 as a query result location.
3. Create databases in the AWS Glue Data Catalog.
4. Create external tables for CSV files.
5. Create external tables for JSON files.
6. Create external tables for Parquet files.
7. Query S3 data using SQL.
8. Query CloudFront access logs.
9. Query Application Load Balancer access logs.
10. Query CloudTrail logs.
11. Create partitioned tables.
12. Optimize queries using Parquet and partition projection.
13. Configure workgroup data usage limits.
14. Export query results to S3.
15. Troubleshoot schema, partition, and data type mismatch errors.

33. AWS Trusted Advisor

1. Access the Trusted Advisor dashboard.
2. Review cost optimization recommendations.
3. Review security recommendations.
4. Review fault tolerance checks.
5. Review performance recommendations.
6. Review service quota checks.
7. Review operational excellence recommendations where available.
8. Identify idle and underutilized resources.
9. Identify underutilized EC2 instances.
10. Review security group configuration recommendations.
11. Configure notifications for Trusted Advisor updates.
12. Create an action plan based on recommendations.

34. AWS Application Migration Service (MGN)

1. Understand the AWS MGN architecture.
2. Initialize the Application Migration Service.
3. Install the AWS Replication Agent on source Linux servers.
4. Install the AWS Replication Agent on source Windows servers.
5. Configure replication settings.
6. Configure staging area subnet and security groups.
7. Monitor source server replication status.
8. Launch test instances.
9. Validate migrated application functionality.
10. Configure cutover settings.
11. Perform final cutover to AWS.
12. Troubleshoot replication and launch failures.

35. AWS Database Migration Service (DMS)

1. Configure a source database endpoint.
2. Configure a target database endpoint.
3. Create a DMS replication instance or supported replication configuration.
4. Configure network connectivity and security groups.
5. Create database migration tasks.
6. Migrate MySQL databases to Amazon RDS.
7. Migrate PostgreSQL databases to Amazon RDS.
8. Perform full-load database migration.
9. Configure Change Data Capture (CDC).
10. Configure ongoing replication.
11. Monitor migration task status.
12. Validate migrated database records.
13. Troubleshoot connection and replication errors.

36. AWS Application Discovery Service

1. Understand agentless and agent-based discovery.
2. Configure Application Discovery Service.
3. Set up AWS Migration Hub.
4. Configure discovery using supported discovery methods.
5. Collect server inventory information.
6. Collect application dependency information.
7. Analyze CPU, memory, and utilization metrics.
8. Group discovered servers into applications.
9. Export discovery data.
10. Analyze application dependencies before migration.
11. Prepare migration assessment reports.

12. Understand the relationship between discovery data and AWS migration planning.

37. AWS DataSync

1. Understand DataSync architecture.
2. Create an AWS DataSync agent where required.
3. Configure an NFS source location.
4. Configure an SMB source location.
5. Configure an S3 destination location.
6. Configure EFS as a destination.
7. Create DataSync tasks.
8. Configure task scheduling.
9. Configure bandwidth limits.
10. Configure include and exclude filters.
11. Perform initial data synchronization.
12. Configure incremental synchronization.
13. Monitor transfer progress and logs.
14. Validate transferred files and permissions.
15. Troubleshoot network and permission errors.

38. Amazon Inspector

1. Enable Amazon Inspector.
2. Enable EC2 vulnerability scanning.
3. Enable ECR container image scanning.
4. Enable Lambda function vulnerability scanning.
5. Configure delegated administration using Organizations.
6. Review vulnerability findings.
7. Identify vulnerable software packages.
8. Analyze CVE and severity information.
9. Prioritize critical and high-risk findings.
10. Integrate Inspector findings with Security Hub.
11. Configure EventBridge notifications.
12. Remediate vulnerabilities and verify updated findings.

39. AWS CodeBuild

1. Create a CodeBuild project.
2. Configure source code from GitHub.
3. Configure source code from CodeCommit where available.
4. Create a `buildspec.yml` file.
5. Configure build environments.
6. Configure environment variables.
7. Build a Java Maven application.
8. Build a Node.js application.

9. Build a Docker image.
10. Push Docker images to ECR.
11. Store build artifacts in S3.
12. Configure build logs using CloudWatch.
13. Configure IAM service roles.
14. Configure build caching.
15. Troubleshoot build failures and dependency issues.

40. AWS CodeDeploy

1. Create a CodeDeploy application.
2. Configure EC2/on-premises deployment groups.
3. Install and configure the CodeDeploy agent on EC2.
4. Create an `appspec.yml` file.
5. Deploy application files to EC2.
6. Configure lifecycle hooks.
7. Deploy applications to Auto Scaling Groups.
8. Configure in-place deployments.
9. Configure blue/green deployments.
10. Integrate CodeDeploy with ALB.
11. Configure deployment rollback.
12. Troubleshoot failed deployment lifecycle events.

41. AWS CodePipeline

1. Create a CodePipeline pipeline.
2. Configure GitHub as the source provider.
3. Configure CodeBuild as the build stage.
4. Configure CodeDeploy as the deployment stage.
5. Configure S3 as an artifact store.
6. Create a CI/CD pipeline for a Java Maven application.
7. Create a CI/CD pipeline for a Docker application.
8. Deploy applications automatically to EC2.
9. Deploy container applications to ECS.
10. Configure pipeline execution triggers.
11. Configure manual approval stages.
12. Configure pipeline notifications.
13. Configure failure handling and rollback.
14. Troubleshoot failed pipeline executions.

42. AWS Health Dashboard

1. Access the AWS Health Dashboard.
2. Review account-specific health events.
3. Review public AWS service health events.

4. Filter events by AWS region and service.
5. Review scheduled maintenance notifications.
6. Review planned lifecycle and infrastructure changes.
7. Configure AWS Health event notifications using EventBridge.
8. Send AWS Health notifications using SNS.
9. Configure organizational health event visibility.
10. Create an operational response workflow for AWS Health events.

43. AWS Cost Explorer

1. Enable Cost Explorer.
2. Review monthly AWS spending.
3. Analyze costs by AWS service.
4. Analyze costs by region.
5. Analyze costs by linked account.
6. Analyze costs using tags.
7. Create cost reports.
8. Compare month-over-month spending.
9. Analyze EC2 and S3 expenses.
10. Configure cost forecasts.
11. Identify high-cost AWS services.
12. Export cost reports for analysis.
13. Analyze Reserved Instance and Savings Plans utilization.

44. AWS Budgets

1. Create a monthly cost budget.
2. Create a usage budget.
3. Create a Savings Plans budget.
4. Create a Reserved Instance budget.
5. Configure actual cost thresholds.
6. Configure forecasted cost thresholds.
7. Configure email notifications.
8. Configure SNS notifications.
9. Configure budget alerts at 50%, 80%, and 100%.
10. Configure budget actions where supported.
11. Monitor account spending against budgets.
12. Create separate budgets for projects and environments.

45. AWS Amplify

1. Create an Amplify application.
2. Connect a GitHub repository.
3. Configure automatic frontend deployments.
4. Deploy a React application.

5. Deploy a Next.js application.

6. Configure build settings using `amplify.yml`.

7. Configure development, staging, and production branches.

8. Configure custom domain names.

9. Configure HTTPS using managed certificates.

10. Configure environment variables.

11. Configure preview deployments.

12. Configure backend resources using Amplify where appropriate.

13. Monitor build and deployment logs.

14. Troubleshoot build failures and domain configuration issues.

46. Amazon Macie

1. Enable Amazon Macie.

2. Configure S3 bucket discovery.

3. Review sensitive data discovery results.

4. Identify publicly accessible S3 buckets.

5. Configure sensitive data discovery jobs.

6. Create custom data identifiers.

7. Detect personally identifiable information (PII).

8. Detect financial and sensitive data patterns.

9. Configure managed data identifiers.

10. Configure Macie findings export.

11. Integrate Macie findings with EventBridge and Security Hub.

12. Configure delegated administration using Organizations.

13. Investigate and remediate sensitive data exposure.

47. Amazon DynamoDB

1. Create DynamoDB tables.

2. Configure partition keys and sort keys.

3. Insert items using the AWS Console.

4. Insert items using AWS CLI.

5. Insert items using Python and boto3.

6. Perform GetItem, PutItem, UpdateItem, and DeleteItem operations.

7. Query and scan DynamoDB tables.

8. Configure Global Secondary Indexes (GSI).

9. Configure Local Secondary Indexes (LSI).

10. Configure on-demand and provisioned capacity modes.

11. Configure auto scaling.

12. Configure DynamoDB Streams.

13. Integrate DynamoDB Streams with Lambda.

14. Configure DynamoDB TTL.

15. Configure Point-in-Time Recovery.
16. Configure global tables for multi-region replication.
17. Configure encryption using KMS.
18. Monitor DynamoDB using CloudWatch.

48. Amazon Aurora

1. Create an Amazon Aurora MySQL cluster.
2. Create an Amazon Aurora PostgreSQL cluster.
3. Configure Aurora DB subnet groups.
4. Configure cluster endpoints.
5. Create Aurora writer and reader instances.
6. Connect to Aurora from EC2.
7. Create databases, tables, and database users.
8. Configure Aurora read replicas.
9. Configure Aurora Auto Scaling for reader instances.
10. Configure automated backups.
11. Create and restore cluster snapshots.
12. Configure Aurora Serverless v2.
13. Configure encryption using KMS.
14. Monitor database performance using CloudWatch and Performance Insights.
15. Configure Aurora Global Database.
16. Troubleshoot connection and database performance issues.

49. Amazon VPC (Virtual Private Cloud)

1. Create a custom VPC.
2. Configure IPv4 CIDR blocks and subnets.
3. Create public and private subnets.
4. Configure Internet Gateways.
5. Configure NAT Gateways.
6. Configure route tables and routes.
7. Associate route tables with subnets.
8. Configure security groups.
9. Configure network ACLs.
10. Create VPC endpoints for S3 and other supported services.
11. Configure VPC peering connections.
12. Configure VPC Flow Logs.
13. Configure Elastic IP addresses.
14. Configure VPC DNS resolution and DNS hostnames.
15. Deploy EC2 instances in public and private subnets.
16. Configure a bastion host for private EC2 access.
17. Configure VPC connectivity using Site-to-Site VPN.

18. Configure VPC traffic monitoring and troubleshooting.
19. Design a three-tier architecture using public, private application, and private database subnets.
20. Troubleshoot routing, NAT, DNS, and security group issues.

50. AWS Transit Gateway

1. Create an AWS Transit Gateway.
2. Create Transit Gateway attachments for multiple VPCs.
3. Attach multiple VPCs to a central Transit Gateway.
4. Configure Transit Gateway route tables.
5. Configure route propagation.
6. Configure route associations.
7. Configure communication between multiple VPCs.
8. Implement centralized network connectivity using a hub-and-spoke architecture.
9. Configure Transit Gateway peering across regions.
10. Configure VPN attachments to Transit Gateway.
11. Configure centralized internet egress architecture.
12. Configure network segmentation using multiple Transit Gateway route tables.
13. Monitor Transit Gateway traffic using Flow Logs.
14. Troubleshoot routing and connectivity issues.